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A MINOR PROJECT PROPOSAL ON

DECENTRALIZED PREPAID SMART POWER GRID

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Abstract

The increasing demand for efficient and transparent electricity management has highlighted the limitations of conventional postpaid and centralized electricity systems. These systems often provide limited real-time information about energy consumption and depend heavily on centralized mechanisms for billing and account management. This project proposes a software-based **Decentralized Prepaid Smart Power Grid** that combines blockchain-based prepaid electricity management, machine learning, simulated energy telemetry, and web-based monitoring within a hybrid system architecture.

The proposed system uses a blockchain smart contract to manage prepaid electricity tokens, user balances, and authorization states. Simulated electricity consumption telemetry is processed through a FastAPI backend and stored in a SQLite database. Machine learning techniques are incorporated to provide intelligent analysis of consumption data, with the Isolation Forest algorithm used for anomaly detection and Linear Regression used for short-term consumption forecasting. The system also estimates the remaining usage duration based on the available prepaid balance and estimated consumption rate. A React-based web interface provides users with information regarding electricity consumption, prepaid balance, forecasted usage, authorization status, and detected anomalies.

The proposed architecture is hybrid in nature. Blockchain provides the decentralized component for prepaid balance and authorization management, while the backend, database, machine-learning module, and frontend provide centralized application services. Physical electricity meters, sensors, relays, and power-distribution hardware are outside the scope of the current project; therefore, electricity consumption is represented through simulated telemetry.

The project is expected to demonstrate the feasibility of integrating blockchain-based prepaid management with machine-learning-based consumption analysis and web-based monitoring. The resulting prototype can serve as a foundation for future integration with physical smart-metering infrastructure, real electrical load control, and deployment on a production blockchain network.

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API : Application Programming Interface

Colab : Colaboratory

CHAPTER 1

INTRODUCTION

1.1 Background

Electricity is an essential resource for households, businesses, and industries, and the increasing demand for electrical energy has created a need for more efficient and transparent methods of electricity management. Conventional electricity billing systems generally operate using postpaid models in which electricity is consumed first and payment is made later. Such systems provide limited awareness of real-time energy consumption and may involve delays in billing, manual intervention, and difficulties in monitoring individual consumption patterns.

The development of smart grid technologies has introduced digital methods for monitoring and managing electricity consumption. Smart grid systems can integrate communication, data processing, monitoring, and automated decision-making to provide improved visibility and control over electricity usage. Prepaid electricity systems are another approach that allows consumers to purchase electricity credit before consumption. By linking electricity usage with an available balance, prepaid systems can encourage consumers to monitor and manage their energy consumption more effectively.

Blockchain technology provides another opportunity for improving the transparency and reliability of prepaid electricity management. A blockchain-based system can maintain transaction records and execute predefined rules through smart contracts. In addition, machine learning techniques can be used to analyze electricity consumption patterns, detect abnormal behavior, and forecast future consumption.

The proposed **Decentralized Prepaid Smart Power Grid** combines these concepts into a software-based system for electricity management. The system uses simulated electricity telemetry, a FastAPI backend, a SQLite database, blockchain-based prepaid balance management, machine learning for anomaly detection and consumption forecasting, and a React-based monitoring dashboard. The system is intended to demonstrate how these technologies can work together to provide a more transparent, intelligent, and automated approach to prepaid electricity management.

The current project focuses on the software architecture and simulation of the proposed system. Physical electricity meters, ESP32 hardware, electrical relays, and actual power distribution infrastructure are outside the scope of the implementation. Instead, simulated telemetry is used to represent the data that would be generated by an edge device in a real-world deployment.

1.2 Gap Identification

Existing electricity management systems address different aspects of energy monitoring and billing, but these functionalities are often implemented separately. Smart metering systems mainly focus on collecting electricity consumption data, while prepaid systems focus on credit-based electricity access and billing. IoT-based energy monitoring systems provide remote access to consumption information, but they may depend on centralized application servers for data processing and account management.

Blockchain-based energy management systems can provide transparent and traceable transactions, but their integration with real-time consumption monitoring and automated electricity management is still an area for further exploration. Similarly, machine learning techniques can be used for detecting unusual energy consumption and predicting future demand, but these capabilities are not always integrated with prepaid electricity management and blockchain-based authorization.

This creates a gap for a unified system that combines prepaid electricity management, consumption monitoring, blockchain-based transaction handling, anomaly detection, and energy forecasting within a single software architecture. The proposed project addresses this gap by integrating these functionalities into a single decentralized prepaid smart power grid prototype.

The system provides simulated telemetry ingestion, prepaid balance management, anomaly detection, energy forecasting, blockchain-based authorization, and a web-based dashboard. By integrating these components, the project demonstrates a software architecture that can potentially be extended to physical smart-metering infrastructure in future work.

1.3 Motivation

The primary motivation for this project is the need for improved control, transparency, and awareness in electricity consumption and prepaid energy management. Conventional postpaid electricity systems do not always provide consumers with immediate information about their current consumption or remaining financial commitment. A prepaid mechanism can provide a more direct relationship between electricity consumption and available credit.

Another motivation is the increasing use of digital technologies for intelligent energy management. Blockchain can provide a transparent mechanism for managing prepaid balances and authorization, while machine learning can help identify unusual consumption patterns and estimate future energy usage. Combining these technologies in a single system provides an opportunity to explore a more intelligent and decentralized approach to electricity management.

The project is also motivated by the need for a practical software prototype that can demonstrate these concepts without requiring physical electrical infrastructure. By using simulated telemetry, the proposed system can be developed and tested in a controlled environment while maintaining an architecture that can potentially support hardware integration in future implementations.

1.4 Objectives

The main objective of this project is to design and develop a software-based **Decentralized Prepaid Smart Power Grid** that integrates prepaid electricity management, blockchain technology, machine learning, and real-time monitoring.

The specific objectives of the project are:

- To design a software architecture for decentralized prepaid electricity management.
- To implement a prepaid electricity mechanism using blockchain and smart contracts.
- To provide a mechanism for recording and managing simulated electricity con-

sumption telemetry.

- To implement automatic prepaid balance deduction based on simulated energy consumption.
- To implement user authorization and cutoff mechanisms using blockchain smart-contract functions.
- To develop an anomaly detection mechanism using the Isolation Forest machine-learning algorithm.
- To develop an electricity consumption forecasting mechanism using linear regression.
- To provide a web-based dashboard for monitoring electricity consumption, prepaid balance, forecasting results, and anomaly information.
- To integrate the backend, database, blockchain, machine-learning components, and frontend into a unified software system.
- To evaluate the proposed system using simulated electricity telemetry without requiring physical electrical hardware.

CHAPTER 2

RELATED THEORY

2.1 Smart Grid

A smart grid is an advanced electricity management system that integrates electrical infrastructure with digital communication, monitoring, data processing, and control technologies. It enables electricity consumption and system conditions to be monitored more efficiently than in conventional power systems. Smart grid technologies can improve energy efficiency, reliability, transparency, and consumer awareness.

The proposed system applies smart grid concepts through a software-based architecture. Rather than directly controlling physical electrical infrastructure, the system processes simulated electricity consumption telemetry and provides mechanisms for prepaid balance management, anomaly detection, consumption forecasting, and authorization control.

2.2 Prepaid Electricity Management

A prepaid electricity system requires consumers to purchase electricity credit before consuming electrical energy. The available credit is reduced according to the amount of electricity consumed. When the available balance becomes insufficient, access to electricity can be restricted according to predefined rules.

Prepaid electricity management provides consumers with greater awareness of their electricity usage and can reduce issues associated with conventional postpaid billing. In the proposed system, prepaid electricity is represented using blockchain-based tokens. Each user is associated with a blockchain wallet address and a corresponding prepaid balance.

2.3 Blockchain Technology

Blockchain is a distributed ledger technology that records transactions in a verifiable and tamper-resistant manner. It provides a mechanism for maintaining transaction records without relying entirely on a single centralized authority.

In the proposed system, blockchain is used for prepaid balance and authorization management. Token purchases, balance deductions, and authorization-related operations are handled through blockchain transactions. This provides traceability for important prepaid operations and forms the decentralized component of the proposed architecture.

2.4 Smart Contracts

A smart contract is a program deployed on a blockchain that defines and executes predefined rules through transactions and function calls. Smart contracts can automate operations and enforce conditions without requiring manual intervention for every transaction.

The proposed system uses a Solidity smart contract named `PrepaidGrid`. The contract provides functions for purchasing prepaid tokens, retrieving balances, checking authorization, deducting balances, and managing anomaly-related cutoff operations.

The major functions of the smart contract include:

- `purchaseTokens()` for purchasing prepaid electricity tokens.
- `getBalance()` for retrieving the current prepaid balance.
- `checkAuthorization()` for checking the authorization state of a user.
- `deductBalance()` for deducting the amount associated with electricity consumption.
- `triggerAnomalyCutoff()` for triggering a cutoff when a significant anomaly is detected.
- `clearAnomalyAndReauthorize()` for restoring authorization after an anomaly has been resolved.

2.5 Blockchain-Based Decentralization

The proposed system incorporates decentralization primarily through the blockchain layer used for prepaid balance and authorization management. Instead of maintaining the complete prepaid state exclusively within the application database, important balance and authorization operations are implemented through a blockchain smart contract.

The `PrepaidGrid` smart contract maintains user-related prepaid balances and authorization states and provides functions for token purchase, balance retrieval, balance deduction, authorization checking, and anomaly-triggered cutoff operations. These operations are recorded as blockchain transactions, providing traceability and reducing dependence on the application database for the authoritative prepaid state.

However, the overall system is not fully decentralized. Telemetry ingestion, data storage, machine-learning processing, and dashboard services are handled by the FastAPI backend and SQLite database. Therefore, the proposed architecture is better described as a *hybrid architecture* that combines centralized application services with blockchain-based decentralized state management.

2.6 Internet of Things

The Internet of Things (IoT) refers to interconnected physical devices that can collect, transmit, and exchange data through communication networks. IoT is widely used in smart energy systems to collect electricity consumption and operational data from distributed devices.

The proposed architecture is designed to support telemetry generated by an edge device. However, physical IoT hardware is not implemented in the current project. Instead, electricity telemetry is simulated and submitted to the backend through an application programming interface. This provides a software representation of the data flow that could be generated by a physical smart-metering device in a future deployment.

2.7 Telemetry and Real-Time Monitoring

Telemetry refers to the collection and transmission of data from a remote source to a processing or monitoring system. In an electricity management system, telemetry may contain information such as power consumption, load percentage, device status, and authorization state.

In the proposed system, simulated telemetry is used to represent electricity consumption data. The backend receives the telemetry through an HTTP endpoint and stores the information in the database. The stored data is then used for monitoring, anomaly detection, forecasting, and dashboard visualization.

2.8 Machine Learning

Machine learning is a branch of artificial intelligence in which computational models learn patterns from data and use those patterns to make predictions or decisions. Machine-learning techniques can be applied to smart energy systems for tasks such as consumption analysis, anomaly detection, and demand forecasting.

The proposed system applies machine learning to two major tasks: anomaly detection using Isolation Forest and electricity consumption forecasting using Linear Regression.

2.9 Anomaly Detection

Anomaly detection is the process of identifying observations that significantly differ from expected or normal behavior. In electricity management systems, unusual consumption patterns may indicate electricity theft, equipment malfunction, abnormal usage, or data inconsistencies.

The proposed system analyzes recent electricity consumption telemetry and identifies observations that deviate from the learned consumption pattern. The detected anomaly information is stored in the database and displayed through the monitoring dashboard. Significant anomalies can also be associated with the blockchain-based cutoff mechanism.

2.10 Isolation Forest

Isolation Forest is an unsupervised machine-learning algorithm designed for detecting anomalous observations. The algorithm isolates observations through recursive random partitioning. Observations that can be isolated using relatively few partitions are generally considered more likely to be anomalous.

In the proposed system, Isolation Forest is applied to a recent collection of simulated electricity consumption samples. The model produces an anomaly result and score for incoming observations. The resulting information is stored in the database and used for anomaly monitoring and further system actions.

2.11 Energy Consumption Forecasting

Energy consumption forecasting is the process of estimating future electricity usage using historical consumption data. Forecasting can provide users with information about expected future demand and can also help estimate the approximate duration for which a prepaid balance may remain sufficient.

The proposed system uses historical electricity consumption data to generate a short-term forecast. The forecast is used to estimate future consumption and calculate an approximate remaining usage duration based on the available prepaid balance and estimated average consumption rate.

2.12 Linear Regression

Linear Regression is a supervised machine-learning technique used to model relationships between variables and predict future values. It estimates a mathematical relationship from historical observations and uses the resulting model to generate predictions.

In the proposed system, Linear Regression is used to estimate future electricity consumption. Historical consumption information is provided to the model, and the resulting prediction is used to generate an hourly consumption forecast. The predicted consumption rate is also used to estimate the approximate remaining usage time associated with the available prepaid balance.

2.13 RESTful API

A RESTful API is an application programming interface that enables communication between software components through standard HTTP operations. REST APIs commonly use methods such as `GET` and `POST` to retrieve and submit information.

The proposed system uses RESTful communication for interaction between its major software components. Simulated telemetry is submitted to the backend using an HTTP request, while other endpoints provide access to balance information, machine-learning results, dashboard data, and system health information.

2.14 FastAPI Backend

FastAPI is a Python-based framework for developing web APIs. In the proposed system, FastAPI acts as the main backend application layer connecting telemetry processing, database management, machine-learning functions, blockchain interaction, and the frontend dashboard.

The backend receives simulated telemetry, stores and retrieves system information, interacts with the blockchain layer, processes machine-learning operations, and provides structured data to the frontend.

2.15 Database Management

A database is used to store, organize, and retrieve information generated by an application. The proposed system uses SQLite for persistent storage of telemetry, user balance information, and anomaly alerts.

The main logical entities maintained by the system are:

- `TelemetryLog`, which stores simulated electricity telemetry and related system status information.
- `UserBalance`, which stores wallet addresses, prepaid balances, authorization state, and related transaction information.
- `AnomalyAlert`, which stores detected anomalies, anomaly scores, severity, resolution state, and associated actions.

2.16 Web-Based Monitoring Dashboard

A web-based monitoring dashboard provides a graphical interface for displaying system information to users. Such dashboards can simplify the interpretation of current measurements, historical trends, warnings, and predictions.

The proposed system uses React with Vite to implement the frontend dashboard. The dashboard displays current load, prepaid balance, estimated remaining time, anomaly score, recent consumption data, forecasted consumption, and recent anomaly alerts. Recharts is used to visualize consumption and forecasting information.

2.17 Software-Based Smart Grid Simulation

Since physical electrical measurement and power-control hardware are outside the scope of the current project, the proposed system uses simulated telemetry to represent electricity consumption.

The simulation provides a controlled environment in which the complete software architecture can be evaluated. It allows the project to demonstrate telemetry processing, database storage, machine-learning analysis, prepaid balance management, blockchain interaction, anomaly handling, and dashboard visualization without requiring physical electrical infrastructure.

2.18 Summary

The concepts presented in this chapter provide the theoretical foundation for the proposed Decentralized Prepaid Smart Power Grid. Smart grid principles provide the overall context, while prepaid electricity management and blockchain-based smart contracts provide the basis for prepaid balance and authorization management. Machine-learning techniques support anomaly detection and electricity consumption forecasting, while RESTful APIs, database management, and web technologies provide the software infrastructure required to integrate these components.

The resulting architecture is a hybrid system in which blockchain provides the decentralized component for prepaid balance and authorization management, while the backend, database, machine-learning engine, and frontend provide centralized application services. The project evaluates this architecture using simulated electricity telemetry rather than physical electrical hardware.

CHAPTER 3

LITERATURE REVIEW

3.1 Smart Grid and Smart Energy Management

Smart grid research has focused on improving the monitoring, communication, and management of electrical energy through the integration of digital technologies. Conventional power systems mainly depend on centralized monitoring and billing mechanisms, whereas smart grid approaches introduce real-time data collection, automated control, and improved interaction between consumers and energy-management systems.

Existing smart energy management approaches have demonstrated the usefulness of digital monitoring for understanding electricity consumption patterns and improving energy efficiency. However, many systems concentrate primarily on monitoring and visualization and do not integrate prepaid balance management, blockchain-based authorization, and machine-learning-based analysis within a single software architecture.

3.2 Smart Metering and Energy Monitoring

Smart metering systems provide electronic measurement and monitoring of electricity consumption. These systems can collect energy-related measurements and transmit them to a central application for analysis and visualization. Real-time monitoring can improve consumer awareness and help identify unusual changes in energy consumption.

Although smart metering provides an important foundation for intelligent energy management, measurement and visualization alone do not provide a complete prepaid electricity-management mechanism. The proposed project therefore extends the monitoring concept by incorporating prepaid balance management, anomaly detection, forecasting, and authorization control into the software architecture.

3.3 Prepaid Electricity Systems

Prepaid electricity systems have been developed as an alternative to conventional post-paid billing. In these systems, consumers obtain electricity credit before consumption, and the credit is reduced according to energy usage. Prepaid systems can improve payment management and encourage consumers to maintain greater awareness of their electricity consumption.

However, traditional prepaid systems generally depend on utility-operated centralized infrastructure to maintain customer balances and manage authorization. This creates a strong dependency on a central service for important account operations. The proposed project explores a blockchain-based approach in which prepaid balance and authorization-related operations are implemented through a smart contract.

3.4 Blockchain in Energy Management

Blockchain technology has been explored in the energy sector for applications such as peer-to-peer energy trading, transaction recording, energy certification, and decentralized energy management. One of the main advantages of blockchain in these applications is the ability to maintain verifiable transaction records and execute predefined rules through smart contracts.

For prepaid electricity management, blockchain can provide a transparent mechanism for recording token purchases and balance-related transactions. However, blockchain-based energy systems may introduce additional complexity related to transaction processing, smart-contract design, and integration with conventional application services.

The proposed project uses blockchain specifically for prepaid balance and authorization management rather than attempting to place the complete application architecture on-chain. This provides a hybrid design in which blockchain handles the decentralized state while conventional software components handle telemetry, data storage, machine learning, and visualization.

3.5 Blockchain-Based Prepaid Energy Management

Research and prototype systems have explored the use of blockchain and smart contracts to manage energy transactions and automate payments. Smart contracts can enforce predefined rules for token transfers, account balances, authorization, and transaction records.

A limitation of many such approaches is that blockchain may be treated primarily as a transaction layer without integration with real-time energy-consumption analysis. The proposed system attempts to connect blockchain-based prepaid management with simulated electricity telemetry, allowing consumption-related information to influence balance deduction and authorization decisions.

3.6 Machine Learning for Electricity Anomaly Detection

Machine learning has been widely investigated for detecting abnormal electricity-consumption behavior. Anomaly detection techniques can be applied to identify patterns that differ from expected usage and may indicate electricity theft, equipment malfunction, faulty sensors, or unusual user behavior.

Both supervised and unsupervised approaches have been studied for electricity anomaly detection. Unsupervised algorithms are particularly useful when labeled examples of theft or faults are limited. Isolation Forest is one such technique that can identify observations that differ significantly from the majority of the data.

The proposed system uses Isolation Forest to analyze recent electricity-consumption telemetry and produce anomaly information. The detected anomaly can be stored, visualized, and associated with the authorization and cutoff functionality of the blockchain layer.

3.7 Machine Learning for Energy Consumption Forecasting

Energy-consumption forecasting has also received significant attention in smart grid research. Forecasting techniques can estimate future electricity usage from historical data and can support planning, demand management, and consumer awareness.

Different forecasting approaches range from statistical methods to machine-learning and deep-learning models. More complex models can provide high predictive capability but may require large datasets and greater computational resources.

The proposed system uses Linear Regression as a comparatively simple forecasting approach. The objective is to demonstrate the integration of consumption forecasting into the prepaid energy-management workflow and to estimate the approximate remaining usage duration from the available balance and predicted consumption rate.

3.8 Web-Based Energy Monitoring Systems

Web-based dashboards are commonly used in energy-management applications to present consumption information, system status, historical trends, and alerts. Visualization can help users interpret electricity-related data and make informed decisions about energy usage.

However, many monitoring dashboards primarily act as visualization layers and do not directly integrate blockchain-based prepaid management and machine-learning results. The proposed system integrates these functions into a single dashboard, allowing users to view consumption, prepaid balance, estimated remaining time, forecasted consumption, and anomaly information together.

3.9 Hybrid Architectures for Smart Energy Applications

A fully decentralized architecture is not always practical for all smart energy-management functions. Data ingestion, machine-learning processing, database operations, and web services are often more efficiently handled using conventional application architectures, while blockchain can be reserved for operations that benefit from distributed verification and transaction traceability.

The proposed system therefore follows a hybrid architecture. The FastAPI backend handles telemetry processing and application logic, SQLite provides local persistent storage, machine-learning components perform anomaly detection and forecasting, and the blockchain layer manages prepaid balance and authorization-related state.

This separation of responsibilities allows the system to combine centralized application services with blockchain-based decentralization without requiring all data and processing operations to be executed on-chain.

3.10 Research Gap

The review of existing approaches indicates that smart metering, prepaid electricity management, blockchain-based energy transactions, machine-learning anomaly detection, energy forecasting, and web-based monitoring have each been studied independently or in different combinations.

However, there remains an opportunity to integrate these capabilities into a single lightweight software prototype. In particular, a system that combines:

- prepaid electricity management through blockchain-based smart contracts,
- simulated real-time electricity telemetry,
- machine-learning-based anomaly detection,
- short-term electricity consumption forecasting,
- authorization and anomaly-triggered cutoff mechanisms, and
- a unified web-based monitoring dashboard

can provide a more comprehensive software demonstration of intelligent prepaid electricity management.

The proposed Decentralized Prepaid Smart Power Grid addresses this gap by integrating these components into a hybrid software architecture. The project does not attempt to replace physical electricity infrastructure. Instead, it demonstrates the interaction among blockchain, backend services, machine learning, database management, and web-based monitoring using simulated electricity telemetry.

3.11 Summary

The reviewed approaches demonstrate that smart metering, prepaid electricity, blockchain, machine learning, and web-based monitoring can each contribute to improved electricity management. Nevertheless, these technologies are often implemented as separate solutions or with limited integration between prepaid management and intelligent consumption analysis.

The proposed project combines these concepts into a unified software prototype. Blockchain is used for decentralized prepaid balance and authorization management, machine learning is used for anomaly detection and consumption forecasting, and conventional backend and frontend technologies are used for telemetry processing, storage, and visualization. This integrated approach forms the basis for the methodology presented in the following chapter.

CHAPTER 4

METHODOLOGY

4.1 Overview

The proposed Decentralized Prepaid Smart Power Grid is developed as a software-based prototype that integrates prepaid electricity management, blockchain technology, machine learning, database management, and a web-based monitoring system.

The project follows a hybrid architecture. Blockchain technology is used for prepaid balance and authorization management, while the backend, database, machine-learning components, and frontend are implemented as conventional software services. Since physical electrical hardware is outside the scope of the project, electricity consumption data is simulated and used as input to the system.

4.2 System Architecture

The proposed system consists of the following major components:

- Simulated electricity telemetry
- FastAPI backend
- SQLite database
- Blockchain smart contract
- Machine-learning module
- Web-based monitoring dashboard

Simulated electricity consumption data is submitted to the backend through a RESTful API. The backend processes and stores the received information and coordinates communication with the machine-learning and blockchain components. The processed information is then made available through the web-based dashboard.

4.3 Telemetry Simulation

Since physical electricity meters and power measurement hardware are not implemented in the current project, electricity consumption is represented through simulated telemetry.

The simulated telemetry contains information such as:

- Device identifier
- Timestamp
- Power consumption
- Load percentage
- Relay status
- Authorization status

The simulated data is submitted to the backend at regular intervals. This allows the proposed system to reproduce the software-level data flow of a smart electricity monitoring system without requiring physical electrical infrastructure.

4.4 Backend Development

The backend is developed using Python and FastAPI. It acts as the primary application layer and provides communication between the telemetry source, database, machine-learning module, blockchain layer, and frontend.

The backend provides interfaces for:

- Receiving and retrieving telemetry data
- Managing prepaid balance information
- Providing machine-learning results
- Providing dashboard information
- Monitoring system health

The backend also communicates with the blockchain through a Web3-based interface to retrieve and update prepaid and authorization-related information.

4.5 Database Management

SQLite is used as the database for storing application data. SQLAlchemy is used to manage interaction between the backend and the database.

The major database entities are:

4.5.1 Telemetry Log

The `TelemetryLog` entity stores simulated electricity telemetry, including power consumption, load percentage, timestamp, relay status, authorization state, anomaly information, and estimated remaining time.

4.5.2 User Balance

The `UserBalance` entity stores information related to users' blockchain wallets, prepaid balances, authorization status, and transaction information.

4.5.3 Anomaly Alert

The `AnomalyAlert` entity stores information related to detected abnormal consumption patterns, including anomaly type, severity, anomaly score, resolution status, and actions taken.

4.6 Blockchain Implementation

The blockchain component is implemented using Solidity and tested using the Hardhat development environment. The prepaid electricity functionality is represented through a smart contract named `PrepaidGrid`.

The smart contract provides functions for:

- Purchasing prepaid tokens
- Retrieving the prepaid balance

- Checking user authorization
- Deducting prepaid balance
- Triggering anomaly-related cutoff operations
- Restoring authorization after an anomaly is resolved

The backend communicates with the smart contract using Web3.py. This provides an interface between the conventional application layer and the blockchain layer.

4.7 Prepaid Balance Management

A user purchases prepaid tokens through the smart contract. The purchased amount is associated with the user's blockchain wallet address.

As simulated electricity consumption is received, the system calculates the corresponding consumption and updates the prepaid balance. The user's authorization state is also checked during the process.

If the available balance becomes insufficient, or if a predefined security condition is satisfied, the system can change the user's authorization state through the smart contract.

4.8 Anomaly Detection

The proposed system uses machine learning to identify abnormal electricity consumption patterns. The Isolation Forest algorithm is used for anomaly detection.

Recent consumption observations are provided to the model. The model evaluates the observations and determines whether a particular consumption pattern differs significantly from the normal pattern.

The anomaly detection process consists of the following steps:

1. Receive simulated electricity telemetry.
2. Extract the relevant consumption information.
3. Maintain a recent set of consumption observations.
4. Apply the Isolation Forest algorithm.

5. Generate the anomaly result and anomaly score.
6. Store the result in the database.
7. Display the detected anomaly through the monitoring interface.
8. Initiate the appropriate authorization action when required.

4.9 Energy Consumption Forecasting

The system uses Linear Regression to estimate future electricity consumption. Historical consumption data is used to train the forecasting model.

The model generates an estimated consumption forecast for the upcoming period. The predicted consumption rate is also used to estimate the approximate amount of time for which the available prepaid balance may remain sufficient.

The estimated remaining time is calculated using:

$$H_{\text{remaining}} = \frac{B}{R_{\text{avg}}} \tag{4.1}$$

where $H_{\text{remaining}}$ represents the estimated remaining time, B represents the available prepaid balance, and R_{avg} represents the estimated average consumption rate.

4.10 Blockchain and Anomaly Management

The proposed system links the anomaly detection process with blockchain-based authorization management.

When an abnormal consumption pattern is detected, the backend records the event and evaluates the associated condition. If the anomaly satisfies the required condition, the backend can invoke the smart contract function responsible for anomaly-related cutoff.

After the anomaly has been investigated or resolved, authorization can be restored through the corresponding smart contract function.

This integration connects machine-learning-based detection with rule-based authorization management.

4.11 Web-Based Monitoring System

A web-based monitoring interface is developed using React and Vite. The interface provides users with information about the current state of the proposed system.

The monitoring interface presents:

- Current electricity load
- Prepaid balance
- Estimated remaining time
- Authorization status
- Anomaly information
- Historical consumption
- Consumption forecast

The frontend obtains the required information from the FastAPI backend through HTTP requests.

4.12 System Workflow

The overall workflow of the proposed system is as follows:

1. Simulated electricity consumption data is generated.
2. The telemetry data is submitted to the FastAPI backend.
3. The backend validates and stores the received data in the SQLite database.
4. The recent consumption data is analyzed by the anomaly detection model.
5. Historical consumption data is processed by the forecasting model.
6. The backend communicates with the blockchain layer to obtain prepaid balance and authorization information.
7. The prepaid balance is updated according to the simulated electricity consumption.
8. If a significant anomaly is detected, the corresponding authorization or cutoff

operation can be initiated.

9. The processed information is provided to the frontend.
10. The web-based interface displays consumption, balance, forecast, and anomaly information to the user.

4.13 Testing and Evaluation

The proposed system is evaluated through software-based testing using simulated electricity telemetry.

The evaluation includes:

- Testing telemetry submission and processing
- Testing database storage and retrieval
- Testing prepaid token purchase and balance retrieval
- Testing prepaid balance deduction
- Testing authorization management
- Testing anomaly detection
- Testing electricity consumption forecasting
- Testing anomaly-triggered cutoff operations
- Testing communication between the backend and blockchain
- Testing the information displayed by the monitoring interface

4.14 Scope of Implementation

The current project focuses on the software implementation and simulation of the proposed Decentralized Prepaid Smart Power Grid.

The implementation includes simulated electricity telemetry, backend processing, database management, blockchain-based prepaid balance management, machine-learning-based anomaly detection, electricity consumption forecasting, and web-based monitoring.

Physical electricity meters, ESP32 devices, sensors, relays, and actual electrical distribution infrastructure are not included in the current implementation. The blockchain component is also evaluated using a local Hardhat development environment rather than a public production blockchain network.

4.15 Summary

The proposed methodology integrates simulated electricity telemetry, backend services, database management, blockchain-based prepaid management, machine-learning techniques, and a web-based monitoring interface into a single software prototype.

The hybrid architecture allows blockchain to handle prepaid balance and authorization-related operations while the backend manages telemetry processing, data storage, machine-learning analysis, and communication with the frontend. The methodology provides a controlled environment for demonstrating the proposed system without requiring physical electrical hardware.

CHAPTER 5

EXPECTED RESULTS

5.1 Overview

The proposed project is expected to result in a functional software prototype of a Decentralized Prepaid Smart Power Grid. The system will integrate prepaid electricity management, blockchain-based authorization, machine-learning-based analysis, database management, and web-based monitoring into a unified software environment.

The expected results of the project are described below.

5.2 Prepaid Electricity Management

The system is expected to provide a functional prepaid electricity management mechanism through a blockchain smart contract. Users will be able to purchase prepaid tokens, retrieve their available balance, and have their balance reduced according to simulated electricity consumption.

The system is also expected to maintain the authorization state of users and provide mechanisms for restricting and restoring authorization based on predefined conditions.

5.3 Blockchain-Based Balance and Authorization

The blockchain component is expected to provide a transparent and traceable mechanism for managing prepaid balances and authorization-related operations.

The smart contract is expected to support token purchase, balance retrieval, balance deduction, authorization checking, and anomaly-related cutoff operations. This will demonstrate the decentralized component of the proposed system.

5.4 Electricity Consumption Monitoring

The proposed system is expected to successfully receive and process simulated electricity telemetry through the backend. The telemetry will provide information related to electricity consumption and system status.

The processed information is expected to be stored in the database and made available for further analysis and visualization.

5.5 Anomaly Detection

The machine-learning module is expected to identify abnormal electricity consumption patterns using the Isolation Forest algorithm.

The system is expected to generate anomaly information and anomaly scores for unusual consumption patterns and store the detected events in the database. The detected anomalies are also expected to be presented through the monitoring interface.

5.6 Electricity Consumption Forecasting

The proposed system is expected to generate short-term electricity consumption forecasts using Linear Regression.

The forecasting module is expected to provide estimated future consumption and an approximate remaining usage duration based on the available prepaid balance and estimated consumption rate.

5.7 Web-Based Monitoring

The system is expected to provide a web-based monitoring interface through which users can observe important system information.

The interface is expected to display:

- Current electricity consumption
- Prepaid balance
- Authorization status
- Estimated remaining time
- Historical consumption information
- Forecasted electricity consumption

- Detected anomaly information

5.8 Integration of System Components

The project is expected to demonstrate successful communication among the major software components, including the backend, database, blockchain, machine-learning module, and frontend.

The integrated system is expected to provide a continuous software workflow from simulated telemetry generation to data processing, anomaly detection, forecasting, prepaid balance management, and user visualization.

5.9 Expected Benefits

The proposed system is expected to provide the following benefits:

- Improved visibility of electricity consumption and prepaid balance.
- Automated management of prepaid electricity credit.
- Transparent and traceable prepaid transactions through blockchain.
- Detection of unusual electricity consumption patterns.
- Estimation of future electricity consumption.
- Improved monitoring through a unified web-based interface.
- A modular software architecture that can be extended in future work.

5.10 Project Deliverables

At the completion of the project, the following deliverables are expected:

- A functional blockchain smart contract for prepaid electricity management.
- A FastAPI-based backend for system communication and data processing.
- A SQLite database for storing telemetry, balance, and anomaly information.
- An anomaly detection module based on Isolation Forest.
- An electricity consumption forecasting module based on Linear Regression.

- A React-based web monitoring interface.
- A software-based simulation environment for evaluating the proposed system.

5.11 Scope of Expected Results

The expected results are limited to the software-based implementation and simulation of the proposed smart power grid. The project is not expected to demonstrate physical electricity measurement, electrical load switching, or actual power distribution.

Physical hardware integration, real electricity metering, deployment on a public blockchain network, and large-scale field testing may be considered as future extensions of the project.

CHAPTER 6

CONCLUSION

The proposed Decentralized Prepaid Smart Power Grid is intended to provide a software-based approach for prepaid electricity management by integrating blockchain technology, machine learning, backend services, database management, and web-based monitoring. The system is designed to address limitations associated with conventional prepaid electricity management by providing improved transparency, automated balance management, consumption analysis, and user monitoring.

The proposed architecture uses a blockchain smart contract to manage prepaid balances and authorization-related operations. The FastAPI backend is responsible for receiving and processing simulated electricity telemetry, managing application data, and communicating with the blockchain and machine-learning components. The Isolation Forest algorithm is proposed for detecting abnormal consumption patterns, while Linear Regression is used for short-term electricity consumption forecasting.

The system follows a hybrid architecture rather than a fully decentralized architecture. Blockchain provides the decentralized component for prepaid balance and authorization management, while the backend, database, machine-learning module, and web interface provide the remaining application services. This separation allows the proposed system to demonstrate the use of decentralized technology without requiring the complete application to operate on the blockchain.

Since physical electricity meters, sensors, relays, and other electrical hardware are outside the scope of the current project, electricity consumption will be represented using simulated telemetry. The project will therefore focus on evaluating the software architecture and the interaction among its major components in a controlled environment.

The proposed system is expected to demonstrate the feasibility of integrating blockchain-based prepaid management with machine-learning-based consumption analysis and web-based monitoring. The developed prototype may provide a foundation for future extensions involving physical smart-metering devices, real electrical load control, broader blockchain deployment, and real-world testing.

Overall, the project aims to demonstrate a practical software architecture for intelligent prepaid electricity management that combines blockchain-based authorization and balance management with automated analysis and monitoring.

REFERENCES

APPENDIX A

APPENDIX B